

Receiver Field of View (FOV)

Theoretical derivation for the TR40 elastic-Mie LiDAR (532 nm, NARIT)

Question

"What is the receiver FOV in milliradians (mrad) for the TR40 LiDAR?"

1. Definition

In a lidar receiver, the **Field of View (FOV)** is the angular acceptance of the complete receiving optical chain — the cone of directions in object space from which incoming light is allowed to reach the detector. Although the optical chain comprises telescope + field stop + collimator + filter + detector, the FOV itself is determined by only two of those elements: the **telescope focal length** f and the **field-stop diameter** D_{FS} placed at the focal plane.

2. Optical derivation

Consider parallel rays from an object at infinity arriving at the telescope at an angle θ from the optical axis. In the paraxial regime they converge to a point in the focal plane at radial distance r from the axis:

$$r = f \cdot \tan \theta \approx f \cdot \theta \text{ (small angle approximation)}$$

A circular aperture of diameter D_{FS} at the focal plane only lets rays through when $r \leq D_{FS}/2$. Solving for the maximum acceptance angle:

$$\theta_{\max} = D_{FS} / (2 f) \text{ (half-angle FOV)}$$

The conventional *full* FOV is twice this value:

$$\text{FOV (full)} = 2 \theta_{\max} = D_{FS} / f$$

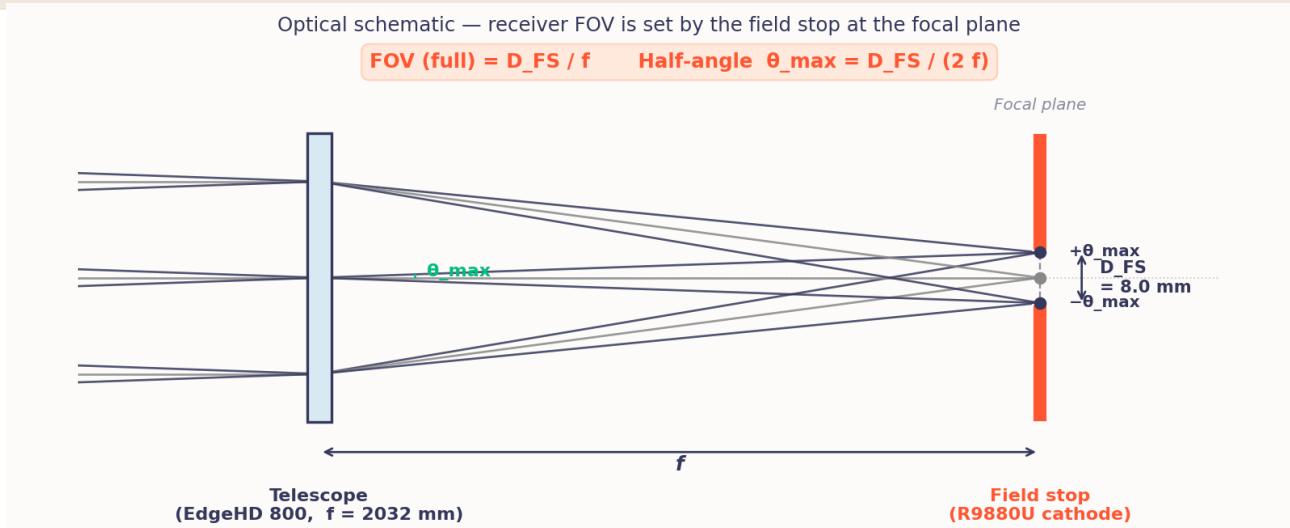


Figure 1 — A field stop of diameter D_{FS} at the focal plane limits the angular acceptance to $\theta_{\max} = D_{FS}/(2f)$ (half-angle).

3. Calculation for the TR40

Substituting the NARIT TR40 hardware values:

Quantity	Symbol	Value	Source
Telescope focal length	f	2032 mm	Celestron EdgeHD 800 (F/10, D = 203 mm)
Field-stop diameter	D _{FS}	8.0 mm	Hamamatsu R9880U PMT photocathode *

* In compact receivers without a dedicated field-stop aperture, the PMT photocathode itself acts as the field stop. If a smaller iris is later added, D_{FS} must be updated accordingly.

Applying the formula:

$$FOV (full) = D_{FS} / f = 8.0 \text{ mm} / 2032 \text{ mm} = 3.937 \times 10^{-3} \text{ rad}$$

Receiver FOV = 3.94 mrad (full angle)

Half-angle θ_{max} = 1.97 mrad

4. Matched-FOV consistency check

A well-designed lidar receiver should have an FOV wider than the laser beam's full divergence so that the entire transmitted beam is captured even with pointing error and thermal drift. The TR40 transmitter has a full divergence of 1.5 mrad:

Parameter	Value	Note
Receiver FOV (full)	3.94 mrad	this work
Laser divergence (full, $D4\sigma$)	1.5 mrad	Quantel VIRON @ 532 nm
Ratio FOV / divergence	2.6 ×	> 2 -> matched-FOV criterion satisfied

The receiver FOV is 2.6 × wider than the laser divergence, comfortably exceeding the common rule-of-thumb of FOV / divergence ≥ 2 (Weitkamp 2005, §5). This margin ensures that the geometric overlap function reaches unity at a reasonable range ($R \approx 142$ m for $O = 0.99$ in the analytical Rice-CDF model) and accommodates small pointing misalignment without losing photons.

References

- 1 Weitkamp C. (ed., 2005). *Lidar: Range-Resolved Optical Remote Sensing of the Atmosphere*. Springer (§5 on geometric overlap & receiver design).
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- 3 Stelmaszczyk K. et al. (2005). Analytical function for lidar geometrical compression form-factor calculations. *Applied Optics*, 44(7), 1323–1331.
- 4 Kumar D., Rocadenbosch F. (2013). Determination of the overlap factor and its enhancement for medium-size tropospheric lidars. *J. Appl. Remote Sens.*, 7, 073591.
- 5 Hamamatsu Photonics K.K. (datasheet). *Metal-Package PMT R9880U series*. Effective photocathode area Ø 8 mm.